Nikhil Patel

Canadian Citizen || nikhil.patel@usask.ca || LinkedIn || Portfolio || GitHub

EDUCATION

University of Saskatchewan, Saskatoon, SK

B.Sc. Electrical Engineering

Relevant Coursework: AutoCAD, Data Structures & Algorithms (Python, UNIX, C), Statistics and Probability (Excel), Signals and Systems (MATLAB), Digital Electronics (KiCAD), Electromagnetism and Waves, Mechanics.

PROJECT WORK

Project Up (Rocketry) – Launch Canada 2025: Propulsion Team Lead | OpenRocket, SolidWorks, ANSYS, Excel. *USST*

- Led the propulsion subsystem for USST's high power rocket "Project UP" by defining performance requirements (thrust, burn time, chamber pressure) and coordinating design, analysis, and testing across the team for competition.
- Utilized CAD based design and iteration of propulsion structures using SolidWorks and Fusion 360, ensuring integration with airframe, recovery, and avionics subsystems, supported by stress, trajectory, and aerodynamic/flow simulations to validate propulsion performance.
- Solely designed a boat-tail component, increasing overall apogee by +15% (~3000 ft) through drag reduction.

Sunrise Simulight | Fusion 360, PCB Design, Electrical Wiring, IOT, BLE, Python Source

- Built an Arduino-based IoT sunrise lamp using recycled LEDs and PWM control to ramp brightness and color temperature for circadian-friendly wake-up lighting.
- Developed embedded firmware and Python configuration tools using UART/BLE and JSON control for scheduling, tuning, and future app integration.
- Designed power electronics with MOSFET dimming, soft-start, and thermal protection, achieving ~85% system efficiency and stable high-current operation. Working towards embedded hardware/PCB design.

Arduino Macropad | Fusion 360, COTS Engineering, Electrical Circuitry, Prototyping, IOT, MATLAB, Source

- Developed Arduino/C++ USB HID firmware for a custom keyboard, implementing multi-profile storage, per-key macros, and configurable debounce for low-latency, cross-platform input automation.
- Designed and integrated electronics hardware including a hot-swappable switch PCB, rotary encoder support, addressable lighting, budgeting for cost efficiency and serviceability.
- Validated system performance through functional testing and deployment, achieving plug-and-play compatibility on Windows/macOS/Linux (standard HID), and rapid assembly via a 3D-printed enclosure and modular BOM.

RELEVANT EXPERIENCE

President / Chief Safety Officer / Rocket Propulsion Lead – USST *Sep 2024 – Current*

University of Saskatchewan Space Design Team (USST).

- **Systems / Project Leadership:** Led multidisciplinary rocketry and CubeSat engineering Design projects.
- **Propulsion / Technical:** Lead the rocketry projects propulsion sub team coordinating propulsion integration with airframe and other subsystems.
- **Systems Administration:** Coordinate with University faculty for facilities access, manage documentation regarding team structure, ratification, and lab use.
- **Financial Management:** Coordinate banking access, cross-checks/reconciliation, and manage sponsor/donor workflows (sponsorship templates, benefit fulfillment, post-competition thank-you packages).

PROFESSIONAL WORK

Summer Camp Counselor - *Southern Alberta Institute of Technology (SAIT).* *Calgary, AB*

- Delivered inclusive, project-based instruction in programming, circuits, 3D modeling, and technology-based learning to students aged 9–17, designing accessible lesson plans and example projects to support diverse learning needs, including students with disabilities.

TECHNICAL SKILLS

CAD & Design: Fusion 360, AutoCAD, OnShape, SolidWorks,
Simulation & Analysis: OpenRocket, SolidWorks FEA, ANSYS, Fusion 360 Simulation,
Embedded & Programming: Arduino variant C/C++, C, MATLAB, Python, HTML5/CSS,
Systems & Platforms: Microsoft Office, Google Suite, Autodesk Online, BLE, IOT, UART, UNIX